Group Q: Phase 2 Report

Research Roots

Peer-review draft

Group Q

Course: 5ARF0 Ethics of AI: Theory and Practice
Eindhoven University of Technology

12 October 2026

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Draft status: The direction described on 12 October focuses on research similarity and distinctiveness, presented in two trees. Section C still describes three trees; Sections B, F and G retain the earlier self-assessment concept. These sections await alignment by their authors. Section H remains incomplete.

Word counts use TeXcount narrative-text counts, excluding headings, citations, references, captions and draft-status notes. Section H contains placeholders rather than completed documentation.

A. Project Title

Research Roots: Finding Related TU/e Research and Reusable Resources

B. Public Summary

Word count: 108/ 150

Our AI-Supported Sustainability Self-Assessment for TU/e Research (ENSU) project explores how researchers can assess the environmental footprint of laboratory or computational experiments. We propose an interactive web application in which a large language model (LLM) helps users describe their activities and develop guidance. A separate step-by-step calculation tool assesses resource consumption, such as equipment runtime and electricity use. The tool could help researchers translate campus sustainability goals into everyday research decisions without adding an institutional approval procedure.

The main anticipated challenges are defining clear assessment boundaries for shared laboratory equipment and infrastructure, ensuring that recommendations can be traced to their sources, and preventing overreliance on automated feedback.

C. Comprehensive Summary

Word count: 1417/ 2000

Problem Definition and Objective(s)

A student starting a new project may find a paper on the same topic yet overlook a useful method or piece of equipment in another TU/e department. The university's research portal already lists projects, publications and equipment (TU/e, n.d.-a). Our project addresses the difficulty of finding relevant work before the student knows the right title, author or search term. The extent of this problem at TU/e has not yet been established and would need to be investigated with researchers.

We propose a website where a student or researcher enters a short description of a planned study: the question, method and, if relevant, equipment needed. The site would show related TU/e work and explain the connection. For example, someone planning a LiDAR (light detection and ranging) experiment might find a similar sensing method used for a different purpose in another department. They could then ask whether a protocol, dataset or setup can be reused. This may save time and resources, but finding a match does not itself save electricity or materials. We would test whether people actually change their plans after using the site.

The main users would be students and researchers at an early planning stage. Project leaders might use it to find partners. Librarians, equipment managers and data stewards would help keep records useful and decide what can be shared. Researchers whose projects appear in the results should be able to decide whether details of unfinished work are visible. The tool would support their judgment; it would not decide whether a project is original or deserves funding. Contacting another group can also create work for that group. The site should help users decide when a conversation is worthwhile rather than send automatic requests to everyone with a similar project.

Background of Current Approaches and Knowledge

The TU/e research portal provides the starting point. We would build on records that the university allows us to reuse rather than create a second collection from scratch. Public indexes such as OpenAlex could give some wider context for published work (OpenAlex, 2026). Neither source, however, tells us about every ongoing project or confidential idea. For an initial version, we would use permitted titles, abstracts and equipment descriptions. A researcher could also choose to add a short summary of work in progress. Every result would show where it came from and when it was last updated.

Existing research gives us a reason to look for connections, but it does not tell us exactly what TU/e researchers need. In one study at a US university, later collaboration was most common when researchers had some topic overlap, rather than almost none or almost complete overlap (Smith et al., 2021). A study including TU/e also argues that sustainability work depends on cooperation across departments and on support from the university (Karo et al., 2026). These findings suggest possible benefits, yet we still need to ask local users when they welcome a match and when it might feel like competition or unwanted exposure.

Scientific-text representation methods provide a technical approach to explore. Models such as SPECTER compare scientific texts using titles, abstracts and citation links (Cohan et al., 2020). Later work shows that performance differs between search and ranking tasks (Singh et al., 2023). We therefore cannot assume that an embedding model, which represents text numerically, will reliably match short project plans. We would compare it with ordinary keyword search and

ask researchers whether the suggested links make sense. We would record our search terms, sources and selection decisions so that the background review can be checked. For example, a sensing method might appear in robotics and building research under different keywords. A method-based link could reveal this, though a researcher still has to decide whether the technique really transfers.

Plan of Approach / Design Strategy

We would use Value Sensitive Design to examine whose needs the site serves and how its features affect stakeholders (Friedman et al., 2008). The initial values are sustainability, privacy, fairness, researchers' control over their ideas and explicability. Explicability requires users to understand why projects are linked. Privacy matters because even a brief description may reveal an unpublished research direction. Fairness matters because departments with more complete records may appear more often in results. These values provide a starting point for stakeholder discussion rather than a final list defined on users' behalf.

First, we would speak with researchers from different departments, a librarian or research-support worker, and a data steward. We would ask how they currently find related work, what makes a match useful, and which details they would be willing to share. A simple mock-up would help them react to something concrete. Their comments would shape the required fields, sharing options and explanations. TU/e requires ethical review for research with participants or personal data, so any interviews or user tests would follow the relevant procedure (TU/e, n.d.-b).

The initial prototype would request a short description of the problem, method and possible application. It would search permitted records by keywords and by scientific-text similarity. Users could indicate whether topic, method or equipment is their priority. Results would be grouped under these headings, with a link back to the original record. We would avoid presenting one exact similarity percentage, since the same number could hide different kinds of connection. If an abstract is missing or a department is poorly covered, the site should say so. We would use a compact search method first and add generated wording only if users understand the results better without extra errors or unreasonable computing costs.

The proposed interface has three linked trees. A *similarity tree* shows which records share a topic, method or resource with the user's idea. A *difference tree* points out how the idea differs from the records found; it cannot prove that the difference is a new research gap. A *future tree* gives questions the researcher could discuss with a supervisor, based on the same records. Each branch should lead to its source. Users could mark a bad match, save a useful one or ask for contact through an approved channel. The tree design is meant to make the information easier to explore; a tree image alone is not evidence of sustainability.

A user's draft description would not automatically join the shared database. They could separately agree to share a short summary, choose who may see it and later withdraw it. TU/e staff would have to approve storage, access and maintenance arrangements under the university's research-data policy (TU/e, n.d.-c). Maintenance would also require responsibility for correcting outdated records and handling complaints about incorrect links. Without this work, even a good search model would soon become less useful.

We would begin with a small set of records from two departments and review missed matches with researchers before expanding it. That would let us check permissions and record quality without assuming that one method works equally well across the university. We would then test the prototype with tasks drawn from real project-planning situations. Users would try both the current portal and the new interface. We would ask domain experts to judge whether the top matches are relevant and whether the explanations are accurate. We would measure time to find a useful record or contact and note where users correct or reject a suggestion. Later, we would ask whether a match led to a conversation, reuse of data or equipment, or a changed plan.

These outcomes are more directly connected to the project objective than click counts. Any estimate of saved resources would need a clear comparison case and would include the site's own computing costs. This is the path by which the tool might help environmental sustainability, but its effect remains to be shown.

Comparison to Alternatives

Our Phase 1 idea was an AI-supported self-assessment of a research activity's environmental footprint. That could be useful once a workflow is defined, but it needs measurements and reference data. The new idea acts earlier, while the researcher can still choose a method or reuse an existing resource. A simpler option would be to improve the current portal's search filters without AI. We would keep that as the baseline; if semantic matching does not help users find better results, the simpler option is preferable.

The main challenges are incomplete records, different terms across fields, confidential work and misleading matches. A polished visualisation could make a weak match appear more certain than the evidence supports. We would show sources and missing information and let researchers correct results. We also need to avoid implying that similar work makes a new project pointless: a similar method can serve a different question. The aim for this phase is a design that can be tested honestly, not a claim that the website has already reduced research waste.

D. Responsible Innovation Project Statement

Word count: 291/ 1000

This statement examines transparency, bias and reliability in the proposed research-matching tool. The measures below address the values identified in our Value Sensitive Design approach: explicability, fairness and researchers' control over results (Friedman et al., 2008).

Transparency

Users submit a research description and receive potentially relevant previous work. They need to understand why each source was selected and be able to check the AI's explanation. We propose the following measures:

- **Source transparency:** Each source should include its title, authors, publication year and a link to the original source, with additional information where available. Researchers can then verify the sources rather than rely solely on the AI.
- **Explanation of relevance:** Each source should have a short explanation of why it was selected, identifying similarities in research questions, methods or other relevant aspects.

Bias and coverage

The tool's results may be biased by the sources available to it. For example, access limited to journal articles or English-language sources could cause researchers to miss relevant previous work. We propose the following measures to reduce this risk:

- **Database coverage:** Users should be able to see which databases the tool searches, so that they understand its limitations.
- **Search strategies:** Where possible, the search should use synonyms, similarities across disciplines and different terminology.
- **User input:** Users should be able to adjust their search query, giving them more control when the AI encounters limitations.

Interpreting results and future work

Our design aims to make results easy to interpret through a chatbot that provides sources and explains their relevance. Future work could help researchers identify ways to improve or differentiate their research. Previous work on the same topic does not necessarily mean that a researcher should abandon a project: there may still be useful differences to explore. In such cases, the tool could also help researchers consider better use of research resources.

E. Reference List

References

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Smith, T. B., Vacca, R., Krenz, T., & McCarty, C. (2021). Great minds think alike, or do they often differ? Research topic overlap and the formation of scientific teams. *Journal of Informetrics, 15*(1), 101104. https://doi.org/10.1016/j.joi.2020.101104

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Eindhoven University of Technology (TU/e). (n.d.-b). *Ethical approval*. https://rdm.tue.nl/docs/before-research/ethical-approval/

Eindhoven University of Technology (TU/e). (n.d.-c). *TU/e RDM framework policy*. https://rdm.tue.nl/docs/intro/tue-policy/

F. User Narrative

Word count: 385/ 500

Dr. Alex, a postdoctoral researcher at Eindhoven University of Technology, is designing a new fabrication run of lithium-ion battery electrode samples within a TU/e laboratory building. Eager to lower total electricity and material waste without changing the scientific parameters that make the batch valid, Alex logs onto the AI-Supported Sustainability Self-Assessment (ENSU) web portal from his laptop or phone.

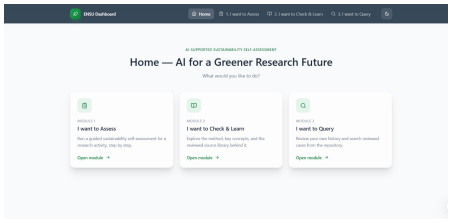
On the home dashboard, Alex selects "1. I want to Assess", initiating a 5-step evaluation process. Rather than selecting rigid checkboxes, Alex inputs a plain-text description: "I make lithium-ion battery electrode samples. I want to minimize material loss and electricity use while preserving our existing electrochemical testing requirements." The language model interprets the description and proposes an activity classification: "This appears to be a laboratory battery-sample activity." Alex clicks "Confirm activity", advancing to the data input sheet.

Alex completes the project fields based on a prior set of data: 6 fabrication attempts yielding 5 valid samples, consuming 60 kWh of meter-tracked laboratory electricity and 1.2 kg of starting materials. The system asks a follow-up question regarding what constitutes a valid sample. Alex types: "A valid sample passes our electrochemical test and safety checks."

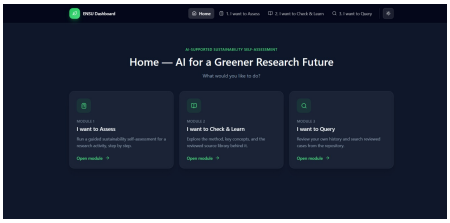
Upon confirming the fields, the tool runs its step-by-step calculation engine and outputs an initial evaluation profile. Rather than showing an aggregated green score or compliance badge, the interface reports separate resource-use indicators: electricity use of 12 kWh per valid sample, which is 2 kWh above Alex's personal sustainability target of 10 kWh. Beneath the values, a targeted suggestion appears: "Pilot a process change that reduces failed samples; record electricity, starting material, valid samples, and test results for the whole batch." Alex clicks "Plan pilot", exits the interface, and tests the proposed process change in the lab.

A week later, Dr. Alex returns to the ENSU web portal. He navigates to "My History" under the "Query" module, and logs the pilot outcomes: 6 attempts, 6 valid samples, 54 kWh, and 1.1 kg of material. The comparison dashboard updates instantly, using visual bars to compare the two batches: batch electricity dropped by 6 kWh, material consumption fell by 0.1 kg, and electricity use reached 9 kWh per valid sample, meeting Dr. Alex's original electricity-use target. Alex saves the session privately to his personal archive, using the recorded comparison to review his research plan.

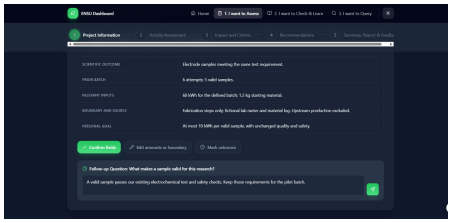
G. Visual Representation of Mock-up



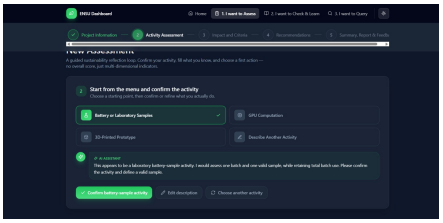
(a) Home page - Light Mode



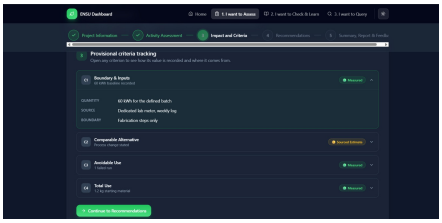
(b) Home page - Dark Mode



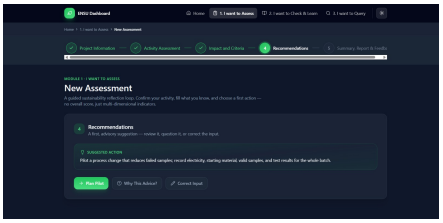
(c) I want to assess - project information



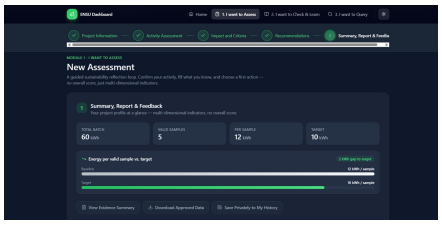
(d) I want to assess - activity assessment



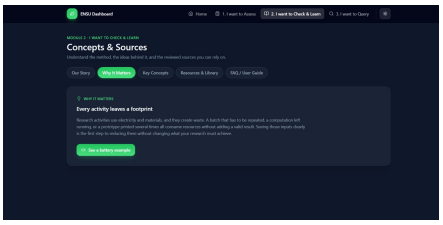
(e) I want to assess - impact and criteria



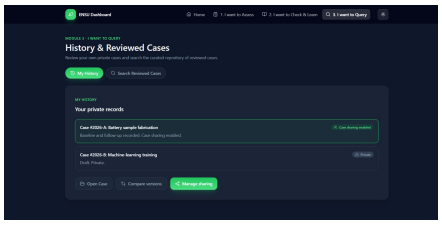
(f) I want to assess - recommendations



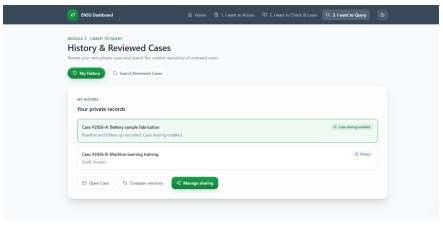
(g) I want to assess - summary, report & feedback



(h) I want to check and learn



(i) I want to query - dark mode



(j) I want to query - light mode

Figure 1: Earlier self-assessment mock-up corresponding to Section F. The example inputs and outputs illustrate the proposed interaction; they are not measured evaluation results.

H. Documentation of Prompts, Sources, and Tools

Provide step-by-step instructions detailing the prompts, database registries, and tools explored, serving as a functional onboarding guide for future project hand-offs.

Prompt Engineering Registry

1. **System Generation Prompts:** ...
2. **Refinement & Interaction Queries:** ...

Technical Infrastructure Setup

[Document technical details like Streamlit/Gradio environments, database stacks, or UI component generators.]